

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol analysis reveals that **port 53 was unreachable**. This conclusion is based on the network traffic analysis, which shows that ICMP **Destination Unreachable** messages were returned with the error message: *"udp port 53 unreachable."*

Port 53 is used for DNS services, and the error message indicates that the DNS request could not be delivered because no service was listening on the destination port. The port number included in the error message is used for query identification.

The most likely issue is that the DNS service was unavailable or blocked, possibly due to a firewall rule or a DNS server misconfiguration, which prevented DNS queries from being processed.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time the incident occurred:

The incident occurred at approximately **1:24 p.m.**, when clients reported that they were unable to access the website.

How the IT team became aware of the incident:

The IT team became aware of the issue after receiving reports from clients stating that they could not access the website.

Actions taken by the IT department:

The IT team attempted to access the website and captured network traffic using a packet analyzer. During this process, DNS queries were sent to the DNS server, and ICMP error messages were received indicating that **UDP port 53 was unreachable**.

Key findings from the investigation:

The DNS server received multiple queries, all of which resulted in ICMP error responses. The following error messages were observed:

- UDP port 53 unreachable (length 254)
- UDP port 53 unreachable (length 320)
- UDP port 53 unreachable (length 150)

Likely cause of the incident:

The most likely cause of the incident is that the DNS service on the destination server was unavailable, stopped, or blocked by a firewall, preventing DNS requests from being successfully processed.

causing all UDP requests to port 53 to fail.